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Terrestrial lidar scanning provides efficient measurements of fire-caused crown scorch in longleaf pine

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Abstract

Background Fire is a dominant ecological process that influences the structure, function, and biodiversity of many ecosystems across the globe. Crown scorch is a primary predictor of tree mortality and is crucial for understanding the mechanisms of tree mortality. Yet, conventional scorch measurements use ocular estimates which are subjective and variable and typically only completed in small areas. Terrestrial lidar systems (TLS) can measure proxies for physiological processes such as leaf water and chlorophyll content and therefore may detect crown injury. Objective measurements of crown scorch that can be rapidly conducted over large areas can advance the mechanistic understanding of tree injury and improve predictions of fire effects.

Results We predicted crown scorch based on patterns of return intensity (range-corrected amplitude) from terrestrial lidar scanning in 253 trees with scorch severity ranging from 0 to 100%. We found that crown scorch consistently altered the empirical distribution of lidar return intensity, particularly in the range of -7.5 to -5 dB, shifting from bimodal to nearly unimodal. Predictions of crown scorch from the best model trained on manually segmented trees were highly correlated with ocular measurements ($R^2 = 0.683$, $RMSE = 0.187$). Predictions from automatically segmented trees were less accurate ($R^2 = 0.460$, $RMSE = 0.227$) than the higher-quality manually segmented dataset primarily due to segmentation errors among small trees (e.g., < 10 cm). To demonstrate the utility of the methodology, we predicted scorch over a 15-ha study area and found the data collection rate was ca. 400 trees h^{-1} of active time—20 times faster than the traditional ocular method. We created an R package CrownScorchTLS to facilitate accessibility of techniques for assessing lidar scorch predictions using the methods herein.

Conclusions Measuring crown scorch with TLS allows accurate and rapid measurements of detailed crown scorch for individual trees at relatively large scales (10s of ha) compared to what is practical with ocular measurements. Moreover, because TLS provides return intensity with high spatial resolution, within-tree measurements of crown scorch may be possible, allowing mapping injury to specific tissues. TLS methodologies and the accompanying R package allow estimates of crown scorch at operational scales, providing much-needed objectivity to improve our mechanistic understanding of the impact of fire on forest ecosystems worldwide.

Keywords Crown injury, Crown scorch, Machine learning, Tree mortality, Prescribed fire, Random forests, CrownScorchTLS

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Resumen

Antecedentes El fuego es un proceso ecológico dominante que influye la estructura, el funcionamiento, y la biodiversidad de muchos ecosistemas alrededor del mundo. El chamuscado que ocurre en las copas de los árboles es un predictor primario de su mortalidad, siendo crucial para entender esos mismos mecanismos de mortalidad. Aún así, las mediciones convencionales del chamuscado de copas usan estimaciones oculares, las cuales son subjetivas y variables y solamente completa en pequeñas áreas. Los sistemas de LIDAR terrestres (TLS) pueden medir substitutos (proxies o aproximaciones) de algunos procesos fisiológicos como el contenido de agua o clorofila de una hoja y por lo tanto podrían detectar heridas en las copas o coronas de árboles. El objetivo de medir el chamuscado de las coronas puede ser rápidamente realizado sobre grandes áreas y avanzar sobre el entendimiento mecanístico del daño a árboles y predecir así los efectos del fuego.

Resultados Predijimos el chamuscado de coronas basados en patrones de la intensidad de retorno (amplitud del rango corregido) mediante el escaneo con LIDAR terrestre de 253 árboles con una severidad de chamuscado que variaba del 0 al 100%. Encontramos que el chamuscado de copas alteraba consistentemente la distribución empírica de la intensidad de retorno del LIDAR, particularmente en el rango de -7,5 a -5 dB, variando de multimodal a casi unimodal. Las predicciones del chamuscado de coronas del mejor modelo entrenado en árboles segmentados manualmente, fueron altamente correlacionadas con las estimaciones oculares ($R^2 = 0,683$; $RMSE = 0,187$). Las predicciones de árboles segmentados de manera automática fueron menos exactas ($R^2 = 0,460$; $RMSE = 0,227$) que las de alta calidad del conjunto de datos manuales segmentados, primariamente debido a errores en la segmentación de árboles pequeños (i. e. < 10 cm). Para demostrar la utilidad de la metodología, predijimos el chamuscado en un área de estudios de 15 ha, y encontramos que el tiempo de colección de datos fue de ca. 400 árboles hr^{-1} de tiempo activo, unas 20 veces más rápido que las estimaciones basadas en el método tradicional de estimación ocular. Creamos un paquete de R (CrownScorchTLS) para facilitar la accesibilidad de técnicas para usar predicciones del chamuscado con LIDAR usando los métodos descritos aquí.

Conclusiones La medición del chamuscado de la corona con TLS permite una medida rápida, exacta y detallada de el chamuscado a nivel de árboles individuales a escalas relativamente grandes (10s de ha) comparado con el tradicional método ocular. Además, dado que la metodología TLS provee de una intensidad de retorno con una gran resolución espacial, la medición de dentro de los propios árboles se hace posible, permitiendo el mapeo de heridas en tejidos específicos. Las metodologías TLS y del paquete estadístico R acompañante permite estimaciones de chamuscado de copas a escala operacional, proveyendo de mucha más objetividad a escala operacional para mejorar nuestro entendimiento mecanístico del impacto del fuego en ecosistemas forestales de todo el mundo.

Background

Fire is a dominant ecosystem process that has important ecological effects across savannas, woodlands, forests, and other terrestrial biomes worldwide (Archibald et al. 2013; Guyette et al. 2012; McLauchlan et al. 2020). Wildland fires are increasingly important as warming climates intensify wildfire regimes and communities struggle to apply prescribed fire effectively (Jolly et al. 2015; Ryan et al. 2013). Fire effects have consequences on carbon dynamics, animal and plant habitat quality, and economic values. Developing a mechanistic understanding of wildland fire effects on woody plants is a key research priority in forested ecosystems (Hood et al. 2018; McLauchlan et al. 2020; O'Brien et al. 2018; Varner et al. 2021). Methods to measure fire effects on woody plants offer promise to advance mechanistic understanding of tree stress and its effects on growth and mortality. Across tree taxa, the primary predictor of post-fire tree death is fire-caused crown injury, apparent after fire as crown “scorch” (Harrington 1993, Hood et al. 2018). For the 38

native tree taxa evaluated by Cansler et al. (2020), all of the best models of post-fire mortality included a crown scorch term; many used crown scorch as the sole predictor. Despite crown scorch's apparent primacy as a driver of tree mortality, conventional measurements of crown scorch are time-consuming and subjective, impeding progress for a mechanistic understanding of fire effects.

The dominant method used to measure crown scorch across studies of post-fire mortality is ocular estimation, but this method has major shortcomings. Ocular estimation employs two trained observers to approximate the proportion of the crown volume browned by the recent fire based on a consensus of their estimations (Cansler et al. 2020; Hood et al. 2018; Peterson 1985). Ocular estimates of crown scorch are subjective and can vary due to variation in lighting, obscuration, irregularly shaped tree crowns, and the ephemeral nature of scorched foliage (sometimes falling within 4–6 weeks of a fire) before eventual senescence (Varner et al. 2021). Ocular estimates have greater error in the intermediate

range (25–75%) compared to the extremes (Hatton et al. 1986), which may obscure critical scorch effects for different species. There is a great need for a more objective method, given the importance of scorch as a primary driver of mortality, and the limitations of field observers to cover large areas over the brief period before senescence when scorch is readily apparent. Measurements of crown scorch that are objective, non-destructive, rapid, and linked to individual trees or specific tissues will expedite research on mechanistic understanding of fire effects and advance frontiers in wildland fire research.

Remote sensing methods can increase the extent and objectivity of scorch measurements, but limitations remain. Previous studies have used high-resolution satellite imagery to measure crown scorch (De Luca et al. 2023; Gibson et al. 2020). This technique is useful for analyzing large areas. Still, the resolution of tree-level (i.e., crown) injury is coarse (e.g., injury classified as scorched vs. intact or low, medium, or high severity). Temporal resolution is dependent on satellite orbits and cloud cover, which can easily exceed the senescence delay in many species (Arnett et al. 2015; Varner et al. 2021). Recent studies using remote sensing from uncrewed aircraft systems (UAS) offer improved spatial, temporal, and spectral resolutions. For example, stereophotogrammetry from UAS has predicted crown scorch heights with high accuracy ($R^2=0.78$) (Arkin et al. 2023; Moran et al. 2022). However, scorching heat originates from flames below tree crowns which scorches trees from the bottom up and disproportionately impacts smaller trees. Thus, imagery acquisition from above tree crowns may obscure the most extreme and ecologically important effects (Arkin et al. 2023). Because of their shorter stature and smaller dimensions, small and midstory trees are more sensitive to crown injury than large trees (Engber and Varner 2012; Hood et al. 2018). Thus, measurements based on above-canopy imagery acquisition may omit measurements of the trees that are most ecologically relevant to studies on injury and mortality.

Terrestrial lidar scanning (TLS) methodologies hold promise for providing objective estimates of crown scorch. Anecdotal evidence suggests that the pattern of lidar reflectance on pine seedlings is modified after scorch, but the method has not been developed (Fig. 4 in Varner et al. 2021). Scorch is broadly defined as a change in foliar color, representing foliage that has suffered heating from surrounding fire. Working hypotheses for the color change (from green to brown) include melting or degradation of the cuticle, internal dehydration, or damage to internal proteins or chloroplasts (Varner et al. 2021). Changes in the cuticle and foliar dehydration are acute, suggesting that it may be possible to use lidar to detect injuries quickly after the fire passes or within a

few days, long before apparent needle-browning. Backscatter from TLS laser pulses (i.e., echo reflectance) provides useful information about the physical properties of objects (Wing et al. 2012; Xu et al. 2017) including leaf moisture content. In one study using a prototype hyperspectral lidar system, Hakala et al. (2015) found that backscatter intensity could be used to estimate chlorophyll concentrations in individual pine crowns or branches. Multi-band lidar sensors or combining lidar scans from sensors operating with different wavelengths has been shown to explain variation in water content as measured by equivalent water thickness (Junttila et al. 2018), or leaf water potential (Junttila et al. 2021). Even single-wavelength lidar instruments have been used to detect chlorophyll concentrations of bananas and leaf water content in horticultural specimens (Saha and Zude-Sasse 2022; Zhu et al. 2017). Further, TLS scans originate beneath crowns where injury is concentrated and may thus provide information on physiological changes to affected tissues with high spatial and temporal resolution.

Here, we develop and evaluate a more objective method to measure post-fire crown scorch in longleaf pine using terrestrial lidar in a recently burned longleaf pine–oak (*Pinus palustris*–*Quercus* spp.) woodland. Our objective was to use field-based ocular estimates of tree-level crown scorch volume (% volume scorch) and use machine learning to develop lidar-derived crown scorch predictions to provide consistent estimates of post-fire crown scorch in longleaf pine. We (1) compared two algorithms for predicting scorch, (2) determined whether scorch could be adequately estimated without the need for pre-burn lidar data collection, (3) assessed the accuracy of measurements collected across a range of tree sizes, and (4) compared tradeoffs among manually and automatically segmented training datasets. Finally, we (5) demonstrated the application of the approach by generating scorch estimates at an operational scale (15 ha).

Methods

We conducted this study in a 15-ha longleaf pine–oak woodland burned approximately biennially congruent with the historic fire return interval (Guyette et al. 2012). Following a prescribed fire, we estimated crown scorch on individual trees using ocular measurements and collected terrestrial lidar scans ~2 weeks before and after burning. We constructed histograms of return intensity for each tree and used them as training data to predict crown scorch. Here, return intensity refers to the instrument's relative reflectance which is given in decibels (dB), the ratio of the range-corrected amplitude of a target to that of a white, flat reference surface (Riegl Laser Measurement Systems GmbH 2017).

Study site and field methods

The study was conducted within the Ichauway Forest Dynamics Plot at the Jones Center at Ichauway located in Baker County in southwestern Georgia, USA (31.268° N, 84.479° W) (Fig. 1). The study area is a longleaf pine–oak woodland, with an overstory dominated by ~100-year-old

longleaf pine and sand post oak (*Q. margarettae*) and a wiregrass (*Aristida beyrichiana*)-dominated understory. Within the 15-ha research plot, all woody stems >1 cm dbh were mapped and censused in 2022, consistent with the protocol of the Smithsonian ForestGEO Network (Anderson-Teixeira et al. 2015). The site has been

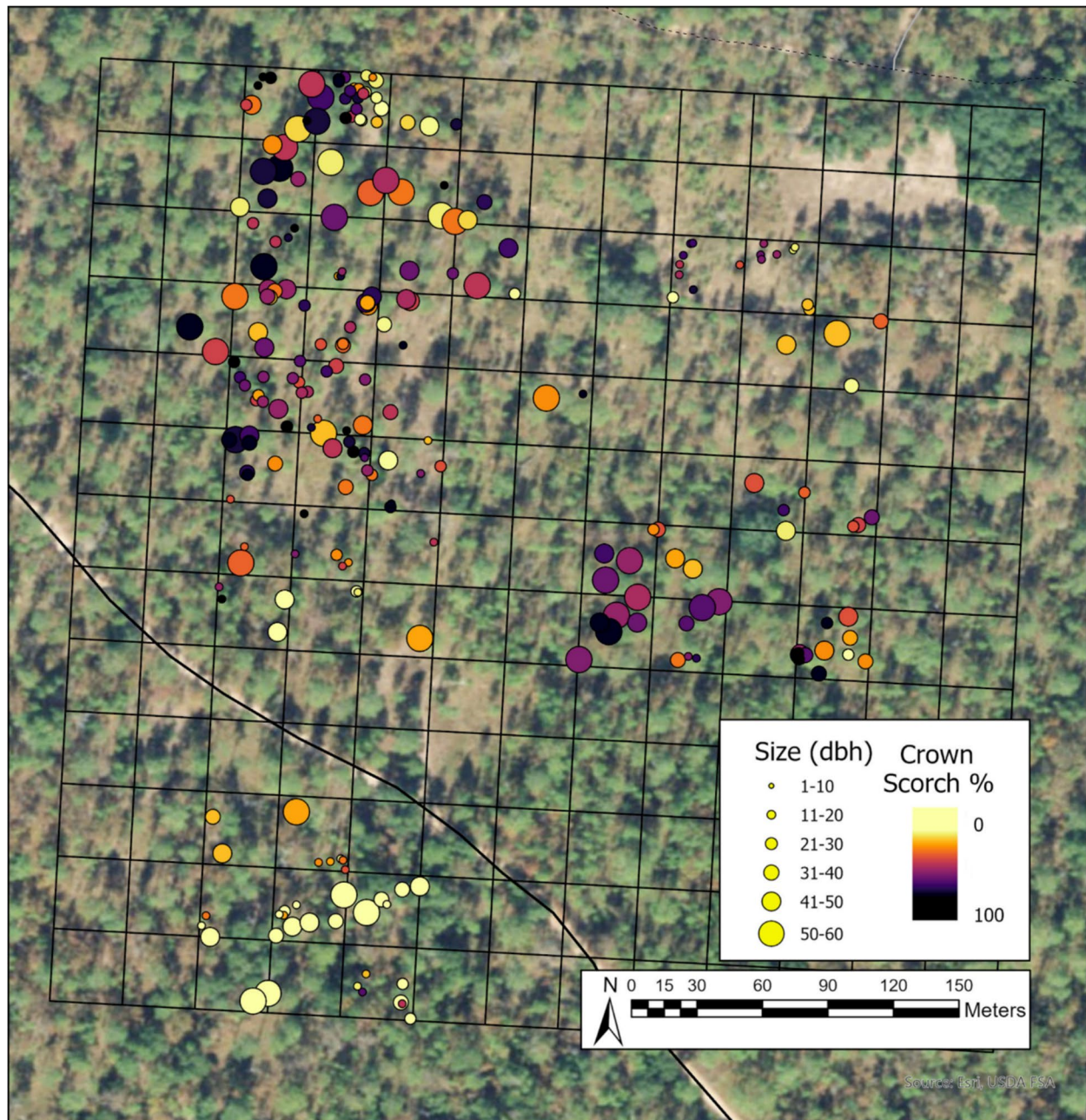


Fig. 1 Ichauway Forest Dynamics Plot encompassing the study area in a longleaf pine–oak woodland in southwestern Georgia, USA (31.268° N, 84.479° W). The 15-ha plot contains over 18,000 mapped trees >1 cm dbh, and colored circles indicate 253 trees selected for analysis spanning a range of tree sizes (proportional to circle size) and crown scorch severity (indicated by color). TLS acquisition occurred at a spacing of 25 m (~250 scan locations)

maintained with regular, frequent prescribed fires for at least a century, although the frequency and seasonality of fires have varied. Since at least 1990, the site has been burned in the late winter–early spring (February–March) on an approximately 2-year return interval (Holland et al. 2019). The climate at the site is humid subtropical with a mean annual temperature of 19 °C and a mean annual precipitation of 1310 mm. The dominant soils in the plot are Lucy, Wagram, and Troup series Ultisols—loamy, kaolinitic sands ranging from well drained to somewhat excessively well drained (NRCS Soil Survey Staff 2023). On 15 March 2023, an operational prescribed fire was ignited within the study area. The unit was ignited primarily using strip head fires. Because a dirt road intersects the study area, the unit south of the road was ignited at 10:45 Eastern Daylight Time (EDT; UTC-4), and the unit north of the road was ignited at 12:15 EDT. The fire was extinct at 12:50 EDT. Flame lengths ranged from 1 to 2 m across most of the unit, typical for burns in these ecosystems. During the fire, ambient air temperatures increased from 9.9 to 13.3 °C, relative humidity declined from 44.6 to 35.4%, and wind speeds varied from 1.6 to 4.8 km h⁻¹.

For convenience, the Ichauway Forest Dynamics Plot is divided into 13 by 13 contiguous tiles, each 0.09 ha (30 by 30 m; Fig. 1). As part of a broader study on tree injury, we selected 47 tiles to monitor crown scorch after prescribed fire. We selected tiles to meet a range of objectives, including some contiguous tiles and tiles that capture variation in species composition and in overstory basal area. Within selected tiles, we measured tree-level crown scorch as a percentage of crown volume for all longleaf pine trees using traditional ocular measurements. From this larger dataset, we classified trees into six 10 cm dbh bins (0–10, 10–20, etc., up to 50–60 cm dbh), and into ten scorch severity classes (0%, 1–10%, 10–20%, etc., up to 90–100%), and we selected 253 trees stratified across the range of sizes and scorch severity (Fig. 1; Table S1). Despite the variety of other hardwood species (e.g., *Quercus* spp.), we included only longleaf pines, as other trees were leaf-off at the time of the burn, precluding visual estimation of canopy scorch.

The observed trees ranged from 2.1 to 58.7 cm dbh. We estimated crown scorch 21–28 days after fire when needles reached peak scorch but before excessive needle fall began. We measured crown scorch using traditional ocular estimates as described in Peterson (1985). We did not differentiate between crown scorch (heat death of foliage and loss of cuticle integrity resulting in brown needles) and crown kill (includes heat death of the meristems and branch cambium) (sensu Varner et al. 2021). In teams of two trained personnel, we conducted ocular estimates of tree-level crown scorch volume as the percent of

total crown with brown needles, observing the full range of each tree's crown (Fig. 2A). Each observer provided an estimate for percent crown volume scorch, and the pair collectively came to a consensus estimate (Peterson 1985).

Lidar scanning and scorch prediction

We acquired terrestrial lidar scans 15 to 20 days before the prescribed burn (24 February–03 March 2023) and again 12–15 days after the prescribed burn (27 March–30 March 2023). We used a RIEGL VZ-400i Terrestrial Laser Scanner (RIEGL LMS GmbH, Horn, Austria) system affixed with an integrated GNSS receiver to capture high-density point clouds within the study area. The scanner uses a short wave near-infrared laser (1510 nm wavelength) with a pulse frequency of 1200 kHz and a nominal 3D position accuracy of 3 mm at 50 m. We spaced scans approximately 25 m apart across the study area including a 15-m buffer. This wide spacing was sufficient given the open canopy and limited midstory maintained by frequent prescribed burning. Following scans, we co-registered and georeferenced point clouds using RiSCAN Pro 2.0 software (RIEGL LMS GmbH, Horn, Austria). We merged and tiled all scans using the landcoults package (Cannon 2023) for R (R Core Team 2023). The resulting point clouds each contained ~5 billion points with an average point density of ~25,000 returns m⁻². To generate a high-quality lidar training dataset, we manually segmented the point clouds of the 253 selected trees pre- and post-burn where scorch was known using the point cloud clipping tools in the software CloudCompare (v2.13.1) (Fig. 2B).

Based on observed variation in return intensity histograms, we tested whether histogram shape predicts scorch severity (Fig. 3). We pre-processed individual tree lidar point clouds in R using the lidR package (Rousset et al. 2020). We segmented and removed tree boles (Fig. 2C) using the stemPoints function in the TreeLS package (de Conto 2020) (Fig. 2C). To represent the distribution of return intensity values for each tree and generate features for machine learning, we constructed normalized histograms for each pre- and post-burn tree crown (Fig. 2D). These histograms capture the shape of the return intensity distribution and varied systematically with the degree of crown scorch (Fig. 3). We binned each histogram into 0.2 dB bins across the observed range (–20 to 0 dB) and used each bin's density as an individual feature to train our random forests models. This approach allowed conversion of continuous intensity data into a consistent set of 100 features suitable for machine learning. To determine the best method for predicting crown scorch, we used two different sets of features: histograms of either raw post-burn intensity values

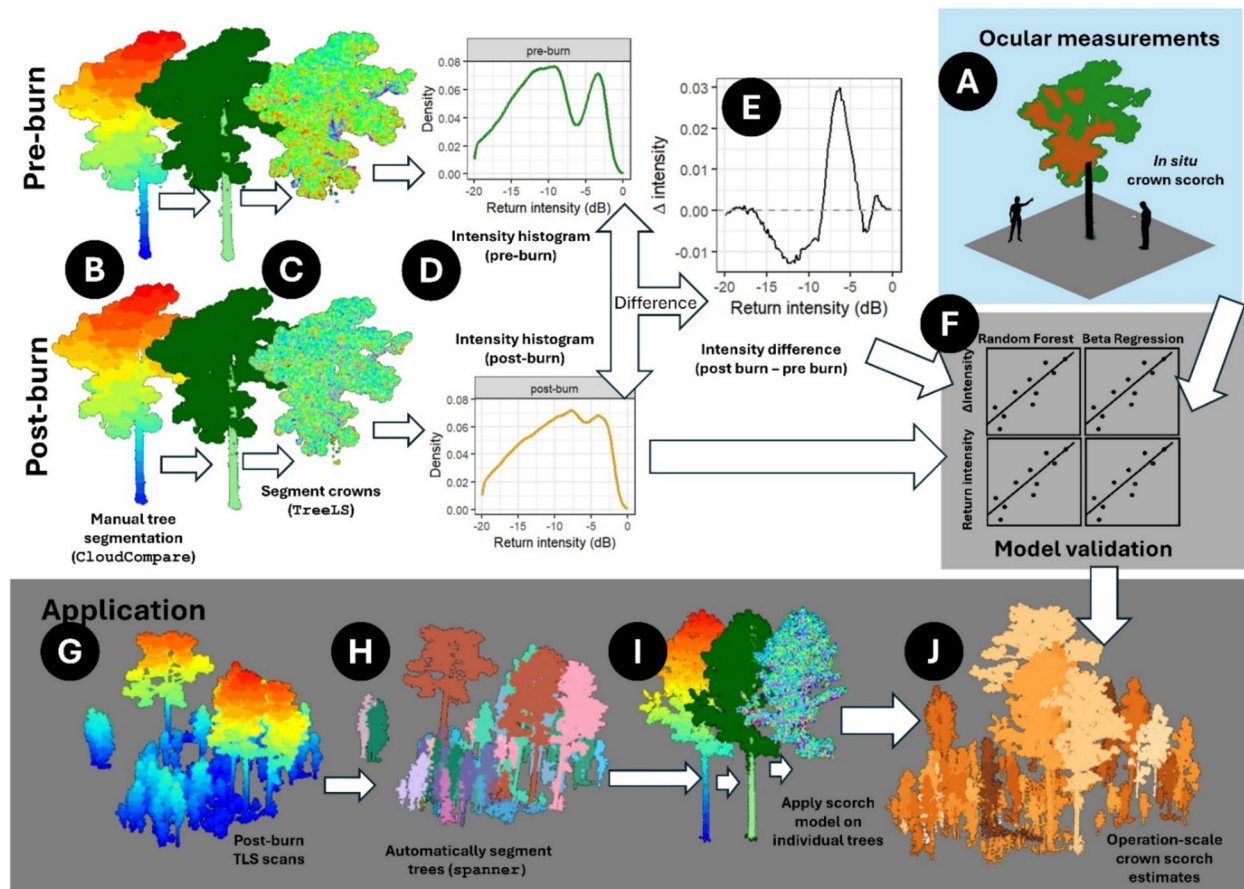


Fig. 2 Schematic methodology for estimating canopy scorch volume using terrestrial lidar scans trained on (A) ocular scorch measurements. We (B) manually segmented pre- and post-burn trees for training, (C) isolated crowns, and (D) generated histograms of return intensity. We (E) calculated Δ intensity from changes in pre- and post-burn histograms, and (F) combined these with training data to model canopy scorch using random forests and beta regression. In application, (G) scanned areas can be (H) automatically segmented, (I) crowns isolated and prediction model applied, resulting in (J) individual crown scorch estimates at an operational scale

(hereafter, intensity) or changes between pre- and post-burn intensity values (hereafter, Δ intensity, Fig. 2E). We crossed this with two statistical prediction methods: random forests or linear modeling (Fig. 2F). For each set of features (intensity vs. Δ intensity), we used Boruta feature selection from the boruta package (Kursa and Rudnicki 2010) to identify the range of histogram values that were important predictors of scorch. Using the randomForest package in R, we constructed 1000 prediction trees using Breiman's random forests algorithm (Breiman 2001; Liaw and Wiener 2002) to predict scorch based on intensity histogram values and Δ intensity histogram values identified as important prediction features. We calculated R^2 for the random forests models from out-of-bag predictions which are generated by aggregating predictions from trees that did not include a given observation in their bootstrap sample. Because scorch values range from 0 to 100%, we also constructed an additive model using a beta regression that took the form:

$$\text{crown scorch} \sim \beta_0 + \beta_1 I_1 + \beta_2 I_2 + \dots + \beta_n I_n \quad (1)$$

where I_n is the intensity histogram value for feature n and β_n is the corresponding model coefficient for feature n . For beta regression models, we randomly assigned 70% of the observations into a training dataset and withheld 30% of the observations for model validation, and we used forward stepwise regression to determine features for inclusion in the model. From the validation data and prediction from each model, we calculated R^2 , mean difference (MD), root mean squared error (RMSE), normalized root mean squared error (nRMSE), and symmetric median absolute percentage error (SMdAPE) using the formula

$$\text{SMdAPE}(\%) = \text{median}\left(\frac{200 \times |p_i - o_i|}{(|p_i| + |o_i|)}\right) \quad (2)$$

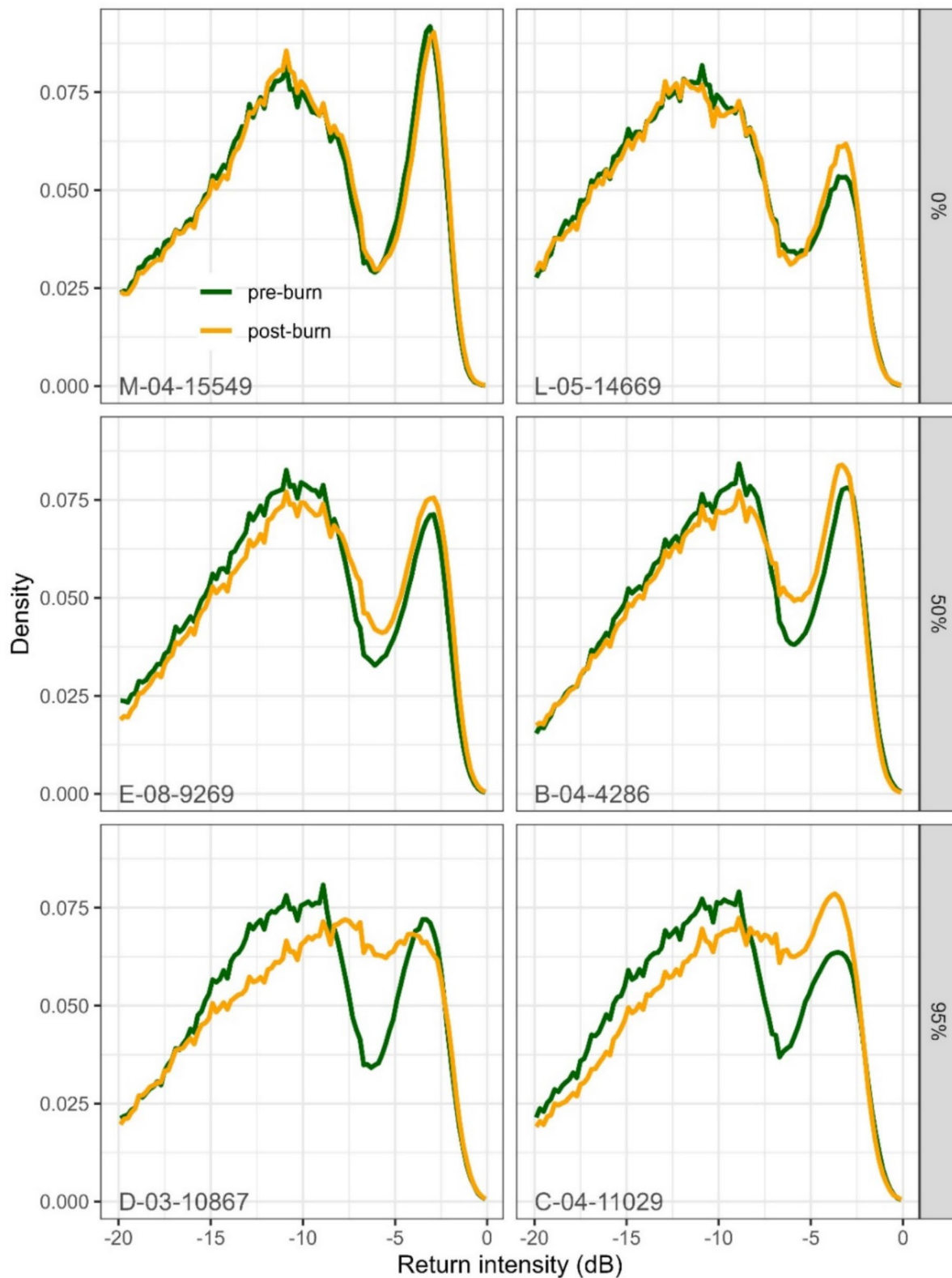


Fig. 3 Histograms of lidar return intensity from six longleaf pines with varying degrees of crown scorch and tree boles removed. Labels correspond to unique tree IDs. Top panel=0% scorch, middle panel=50% crown scorch, and bottom panel=95% crown scorch, according to field-based ocular measurements. Lines indicate return intensity histogram before (green) and after (yellow) a prescribed fire. Note increasing departure between pre- and post-burn intensity histograms for trees after severe scorch (e.g., >75%), especially within the range of -15 and -5.5 dB

where p_i and o_i represent the predicted and observed canopy scorch values, respectively. We compared models based on these evaluation metrics and changes in these measures of accuracy across various tree size classes.

Applications

After determining that statistical models satisfactorily predicted crown scorch, we sought to evaluate the application of our approach in several ways. (1) We evaluated the utility of the method using automated tree segmentation rather than manual segmentation. (2) We applied lidar-derived crown scorch predictions to a 15-ha research plot to demonstrate the method across a large, forested area. (3) We adapted the method to estimate and map crown scorch within individual tree crowns.

Automated methods for segmenting trees may be preferred for efficiency when manual segmentation is impractical. Using the post-burn lidar scene, we automatically segmented the 253 trees using the `segment_graph` function in the `spanner` package for R (Sánchez Meador and Donager 2021), which automatically extracts point clouds for individual trees using a graph theory approach. After determining that the random forests model trained on intensity histograms was the best model, we constructed a new model from automatically segmented trees and their associated ocular scorch measurements. Visual inspection suggested that small trees may have fewer lidar returns or be affected by segmentation errors, which can reduce model performance. To quantify this effect, we evaluated whether excluding small trees from either training dataset improved model accuracy as determined by R^2 . For each dataset, we built a series of random forests models, progressively excluding trees smaller than a minimum diameter threshold, beginning at 2.5 cm dbh and increasing in 2.5 cm increments up to 45 cm at which point further increases reduced the training set to fewer than 30 individuals.

Second, to demonstrate the scorch prediction method across a large forested area, we mapped and segmented over 18,000 individual trees from post-burn lidar scans of the 15-ha Ichauway Forest Dynamics Plot (Fig. 2G) using the `segment_graph` function in the `spanner` package for R (Sánchez Meador and Donager 2021) (Fig. 2H) including 6111 longleaf pines. For pines, we generated intensity histograms as above and used the best model of scorch prediction to estimate crown scorch (Fig. 2I). We linked crown scorch predictions to individual trees and visualized these for a 30 m by 60 m transect (0.18 ha) and a 120 m by 120 m plot (1.44 ha; Fig. 2J), and the entire study area (15 ha). We assessed how crown scorch varied with dbh to demonstrate typical analytical applications. We also constructed a semivariogram of crown scorch to assess spatial patterns of scorch severity for lag distances

with a 2-m interval, and we fit a variogram by applying an exponential model using the `gstat` package (Pebesma 2004) in R.

Lastly, to explore the potential for mapping scorch within individual tree crowns, we conducted a proof-of-concept analysis following a separate prescribed fire on 1 May 2024. We selected and photographed three longleaf pine trees between 10–13 cm dbh and 5–7 m tall and ocularly estimated crown scorch for each. For each tree, we segmented the lidar point cloud and removed the tree bole using the `stemPoints` function of `TreeLS` (de Conto 2020). Then, we used the `point_metrics` function in the `lidR` package (Roussel et al. 2020) to estimate the proportion of returns with values within the range of -7.5 to -5.5 dB within a neighborhood radius of 15 cm. We chose this range as it was important in the Boruta feature selection. Using the field-derived ocular measurement of scorch, we classified the highest x percentile of returns as scorched, where x is the field-estimated value of total tree scorch. We present these results alongside tree photographs for visual comparison.

Results

Lidar-derived crown scorch estimates strongly agreed with traditional ocular measurements. Our field dataset included 253 *Pinus palustris* trees, ranging in size from 2.1 to 58.7 cm dbh, and ocular estimates of crown scorch ranged from 0 to 100%. Histograms of return intensity showed distinct patterns that differed between pre-burn and post-burn trees (Figs. 3, 4). Pre-burn return intensity histograms had a bimodal distribution, with a trough between approximately -7.5 and -10 dB. Histograms of return intensity for post-burn scans also showed a bimodal distribution, but the trough between -7.5 and -10 dB was less pronounced than in pre-burn scans. For trees with low scorch (e.g., $<25\%$), the difference in histograms of return intensity was negligible (Fig. 4C, red line). However, for trees with higher scorch, histograms of return intensity from post-burn scans showed a large departure from those of pre-burn scans, with a slight decrease around -15 dB and a large increase around -7.5 dB (Fig. 4B, C). The Boruta feature importance analysis indicated that histogram features between -7.5 and -5.5 dB were especially important for predicting crown scorch from histograms of raw return intensity and for Δ return intensity (Fig. S1).

All models were significant predictors of crown scorch (all $p < 0.001$). In general, models using raw return intensity values performed better than those based on Δ return intensity. The random forests model on histograms of raw return intensity values was the best model with superior values across all error metrics and had the highest agreement with field-measured values ($R^2 = 0.683$;

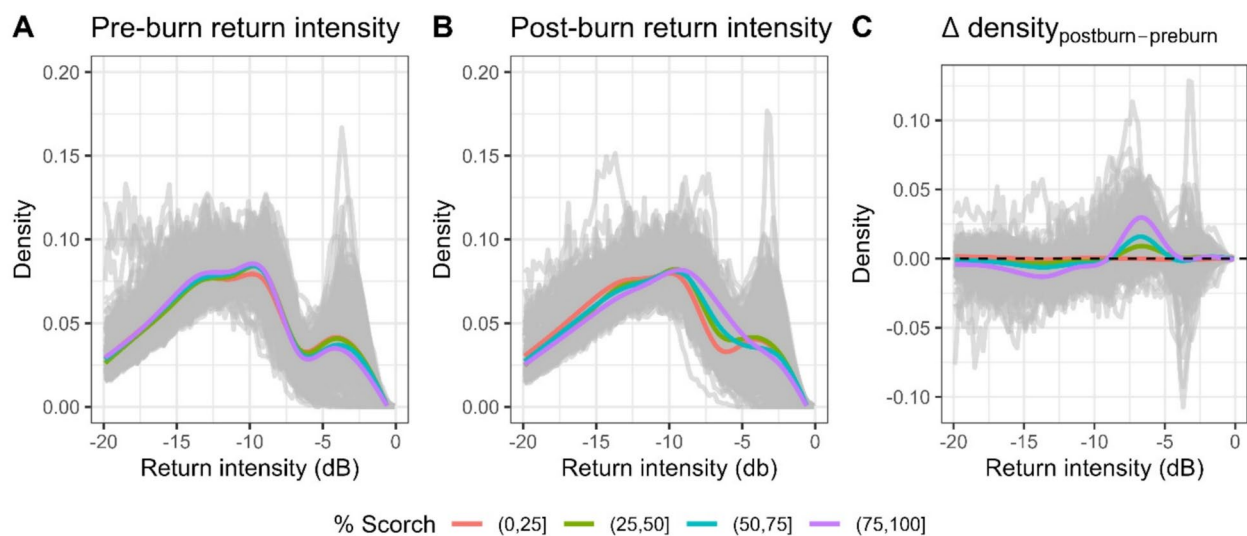


Fig. 4 Return intensity histograms from 253 scanned longleaf pines (A) before prescribed fire, (B) after prescribed fire, and (C) differences in return intensity histogram values among individual trees before and after fire. Smoothing function applied separately for four levels of scorch using a loess function to illustrate divergence in pre- and post-burn intensity for higher levels of crown scorch

RMSE=0.187), lowest MD, RMSE, nRMSE, and second lowest SMdAPE (Fig. 5, Table 1). The second-best model was the beta regression model on histograms of raw return intensity values ($R^2=0.660$, RMSE=0.194). By comparison, the models of scorch based on Δ return intensity performed worse for both random forests ($R^2=0.625$, RMSE=0.204) and for beta regression ($R^2=0.538$, RMSE=0.232). In general, all models slightly overpredicted crown scorch at low values and slightly underestimated scorch at high values, though the random forests model had the lowest bias as measured by mean differences (<0.0001) while the Δ intensity model using beta regression had the highest bias (−1.828) (Fig. 5, Table 1).

The accuracy of terrestrial lidar-derived estimates of crown scorch decreased with tree size (Fig. 6). For the best models of raw intensity, R^2 values were highest in the largest size classes (e.g., >40 cm, Fig. 6, Fig. S4), and RMSE was lower (Fig. 6, Fig. S4). Values of R^2 were greater than 0.81 for the largest size classes (50–60 cm dbh) and decreased to less than 0.54 for the smallest size classes (0–10 cm dbh). Similarly, RMSE was less than 0.14 for the largest size classes (>40 cm dbh) and increased to greater than 0.22 for the smallest size class (<10 cm dbh, Fig. 6, Fig. S4).

Compared to the model built from manually segmented trees, the random forests model based on intensity returns performed worse when trained with automatically segmented trees (Fig. 7). We found that predicted scorch from this model had an R^2 of 0.460 and RMSE of 0.227. Excluding small trees from training

datasets improved model performance for both manual and automated segmentations (Fig. 8). For manually segmented trees, removing trees <10 cm dbh increased R^2 from 0.683 to 0.722. For automatically segmented trees, R^2 improved from 0.460 to 0.574 when trees <10 cm were removed from training data. In both cases, model accuracy improved further as progressively larger size thresholds were excluded (Fig. 8).

To demonstrate the applicability of this approach for larger study areas, we applied the best model across all error metrics (random forests model of post-burn return intensity values) to predict crown scorch across all longleaf pines in the 15-ha study area (Figs. 9, 10, 11A). We found that estimated crown scorch was highest among trees that were 1–20 cm dbh (Fig. 11B), and that spatial autocorrelation of crown scorch was limited to ~50 m based on a semivariogram of crown scorch (Fig. 11C).

Discussion

Our methodology provides the first evaluation of terrestrial lidar for post-fire scorch estimation. Our method produced accurate, tree-level predictions of scorch using machine learning trained on ocular data (Fig. 6). TLS enabled rapid wall-to-wall assessment during the narrow window prior to foliage senescence (Figs. 9, 10, 11). To support the adoption of the TLS method, we provide an R package—CrownScorchTLS—implementing these methods: <https://github.com/jbcannon/CrownScorchTLS>.

We found that histograms of lidar return intensity provide a reliable proxy for estimating canopy scorch.

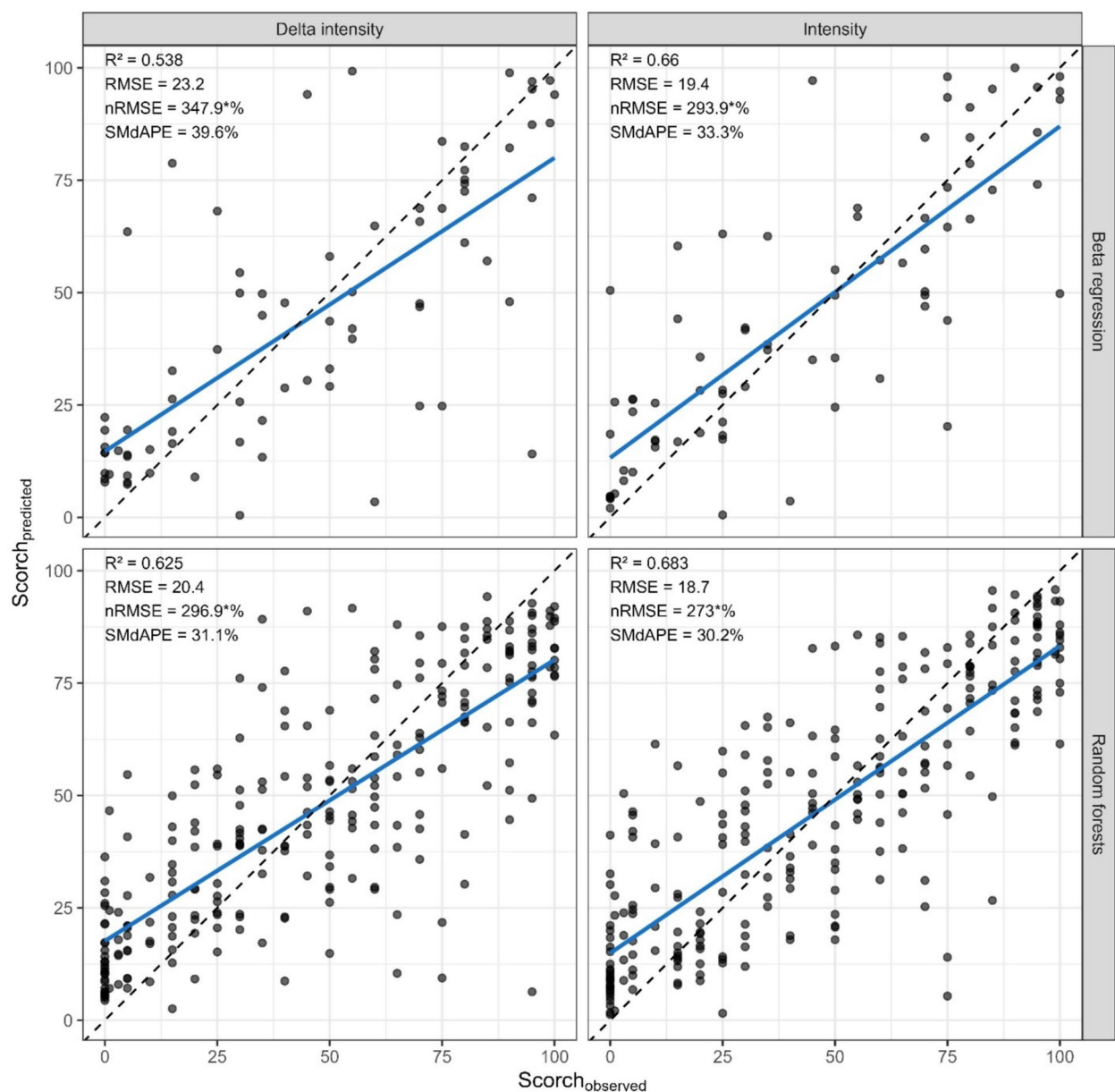


Fig. 5 Model fits comparing four methods for predicting post-fire crown scorch on 253 longleaf pines including (top left) a beta regression model using difference between intensities between pre- and post-burn scans (Δ intensity); (top right) a beta regression using raw intensity of post-burn scans; (bottom left) a random forests model using (Δ intensity); and (bottom right) a random forests model using raw intensity of post-burn scans. Blue line represents model fit of predicted versus observed data. Dashed line represents 1:1 line

Reflectance of lidar pulses is attenuated by several factors, including travel distance (range) and incidence angle (Xu et al. 2017), but it can also be impacted by vegetation factors such as moisture and chlorophyll content (Saha and Zude-Sasse 2022; Zhu et al. 2017). Histograms of intact trees exhibit a distinctly bimodal distribution with a broad peak between -20 and -5 dB which is indicative of vegetation and a narrow peak between -5 and 0 dB which mostly consists of bark and branches (Fig. 3). This

difference is leveraged in segmentation algorithms to isolate leaves from woody tissue (e.g., Béland et al. 2014; Wing et al. 2012). We used range-corrected reflectance values, so the observed peak separation is due to differences in physical and structural properties of leaves and branches. The broader peak for vegetation is attributable to their narrow shape and more complex structure which consists of a wide range of leaf orientations and incidence angles (Béland et al. 2014). After canopy scorch, tissue

Table 1 Model evaluation metrics for four methods of predicting post-fire crown scorch from 253 manually segmented (Man.) longleaf pine tree point clouds and one method using automated segmentation (Auto.). Accuracy metrics include mean difference (MD), root mean square error (RMSE), normalized RMSE (nRMSE), and symmetric median absolute percentage error (SMdAPE). Row in bold represents the best model based on all metrics of fit.

Response	Model	Segment	R^2	MD	RMSE	nRMSE	SMdAPE
Intensity	Beta regression	Man.	0.660	−1.828	0.194	44.6	33.3
Intensity	Random forests	Man.	0.683	<0.001	0.187	39.8	30.2
Δ intensity	Beta regression	Man.	0.538	0.776	0.232	52.1	39.6
Δ intensity	Random forests	Man.	0.625	<0.001	0.204	43.2	31.1
Intensity	Random forests	Auto.	0.460	<0.001	0.227	43.3	28.8

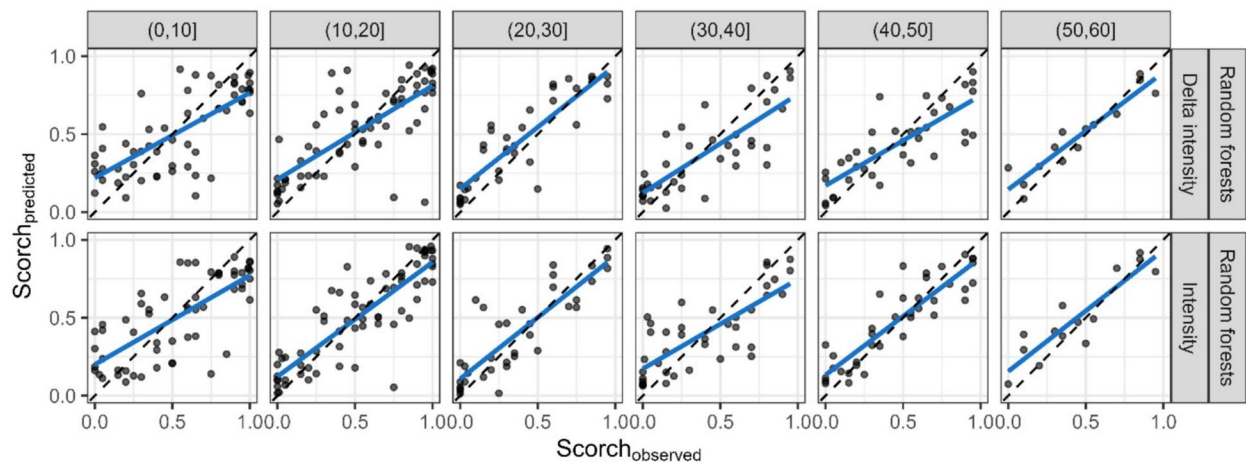


Fig. 6 Comparison of canopy scorch volume as predicted by random forests model ($\text{Scorch}_{\text{predicted}}$) to field-observed ocular estimates ($\text{Scorch}_{\text{observed}}$) for trees included in the study across six size classes from 0 to 60 cm dbh. Predictions based on leave one out cross-validation. Blue line represents model fit of predicted versus observed data, and dashed line represents 1:1 line

necrosis can result in physical changes in vegetation such as cuticle rupture, moisture loss, and chlorophyll denaturation (Varner et al. 2021), which alter infrared absorption and change lidar reflectance. Our results suggest that scorched needles have increased return intensity, causing a rightward shift of vegetation returns in the intensity histograms (Fig. 4). Necrotic needles maintain their complex physical arrangement, and thus still have a lower intensity than that of trunk and branches, resulting in return intensities that are intermediate between the peaks for green vegetation and bark or branches, as seen in Fig. 4. Both the feature importance analysis (Fig. S1) and the change between pre- and post-burn histograms (Fig. 4C) suggest that this peak is most evident between -7 dB and -5 dB.

Among the prediction models compared, we found that the random forests model using raw intensity data performed best ($R^2=0.683$), having the lowest values for most error metrics (Table 1). The beta regression model trained with differences between pre- and post-treatment histograms performed slightly worse ($R^2=0.660$). This

result suggests that only a post-fire survey is required for accurate scorch measurements, making the method useful for both planned prescribed fires and unplanned wildfires where pre-burn measurements would usually be impossible. All models tended to overpredict lower values of scorch and underpredict higher values of scorch (Figs. 5, 6), which is expected behavior of random forests models due to their ensemble approach with a tendency to regress predictions toward the mean (Zhang and Lu 2012). This effect was more apparent in smaller trees (e.g., <30 cm dbh) compared to larger trees, where predictions were improved (Fig. 6, Fig. S4C, D).

Our measurements of scorch obtained by TLS compare favorably to those collected by UAV systems. Arkin et al. (2023) found strong agreement between field-collected measurements of crown scorch and those from UAV stereophotogrammetry ($R^2=0.84$), which was comparable to our measurements of large individuals. However, the use of UAV systems may miss ecologically relevant components, such as understory and midstory individuals (e.g., <10 cm dbh) and the underside of large canopy

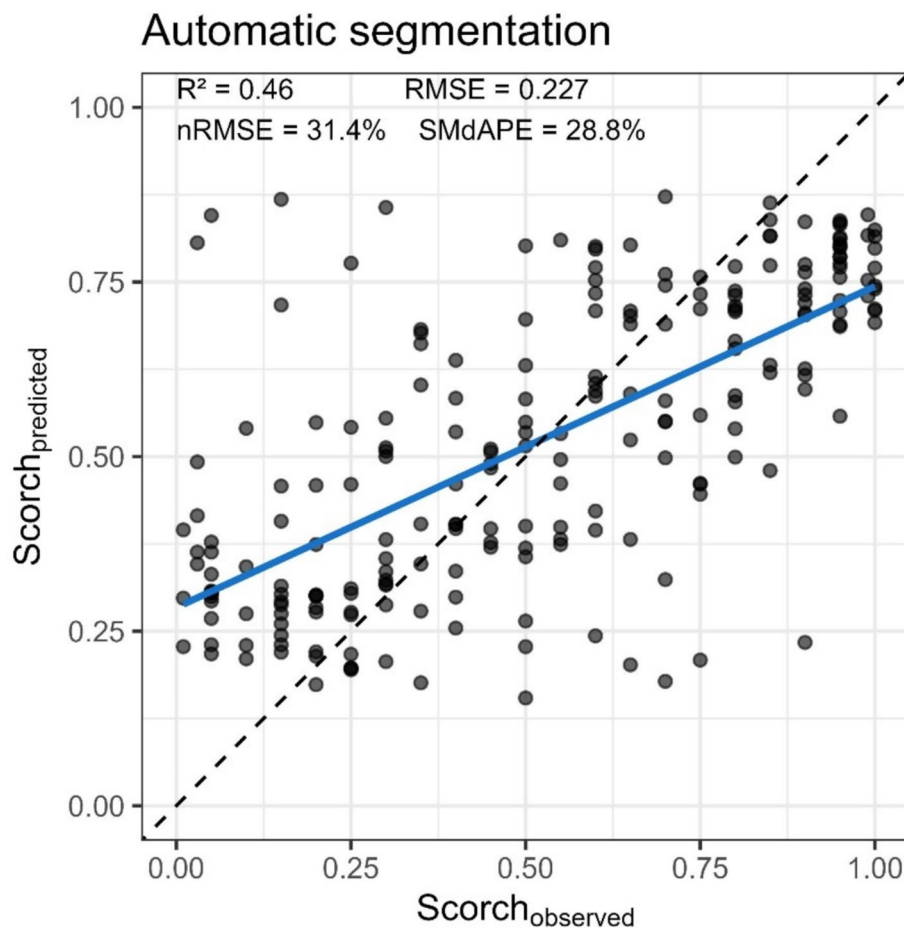


Fig. 7 Random forests model using raw intensity of post-burn scans using a training set generated from automatically segmented trees. Blue line represents model fit of predicted versus observed data. Dashed line represents 1:1 line

crowns where heat effects may be most pronounced, that can be documented with TLS methods (Fig. 11). Moreover, the high spatial resolution may allow inference of physiological changes in specific tissues relevant for advancing the understanding of mechanisms of fire injury (Fig. 12).

Despite being a reliable proxy for crown scorch overall, trees greater than 40 cm had more accurate predictions of scorch ($R^2 > 0.80$) relative to smaller trees. One possible explanation for this difference is that the point clouds of the largest trees had a higher number of lidar returns by several orders of magnitude. Because our prediction model relies on the shape of return intensity histograms, those constructed from fewer points may be more heavily influenced by noise that masks subtle changes in the return intensity distribution. Trees <10 cm dbh averaged ~60,000 returns per tree, whereas the largest trees in our sample comprised over 3 million returns per tree (Fig. S2). We conducted ocular measurements 21–28 days after the burn, when canopy scorch was most clearly

expressed. We collected lidar scans about 10 days earlier, before scorch reached its peak visual clarity. Despite this earlier timing, our lidar-based predictions remained reasonably accurate, suggesting that TLS can detect scorch even before it reaches maximum visual contrast. These results also suggest that collecting scans closer to peak expression could further improve predictive accuracy.

We found that models trained using manually segmented data ($R^2 = 0.683$) outperformed models trained using automatically segmented trees ($R^2 = 0.460$). Although manual segmentation is more labor-intensive, it yields a high-quality training dataset that largely eliminates segmentation errors and can be reused for future investigations employing similar methodologies in similar ecosystems. While less accurate, automated segmentation required less labor and provided reasonably accurate scorch predictions, making it a practical option when greater efficiency is needed. Regardless of the segmentation approach, model performance improved when small trees (e.g., <10 cm dbh) were excluded from the

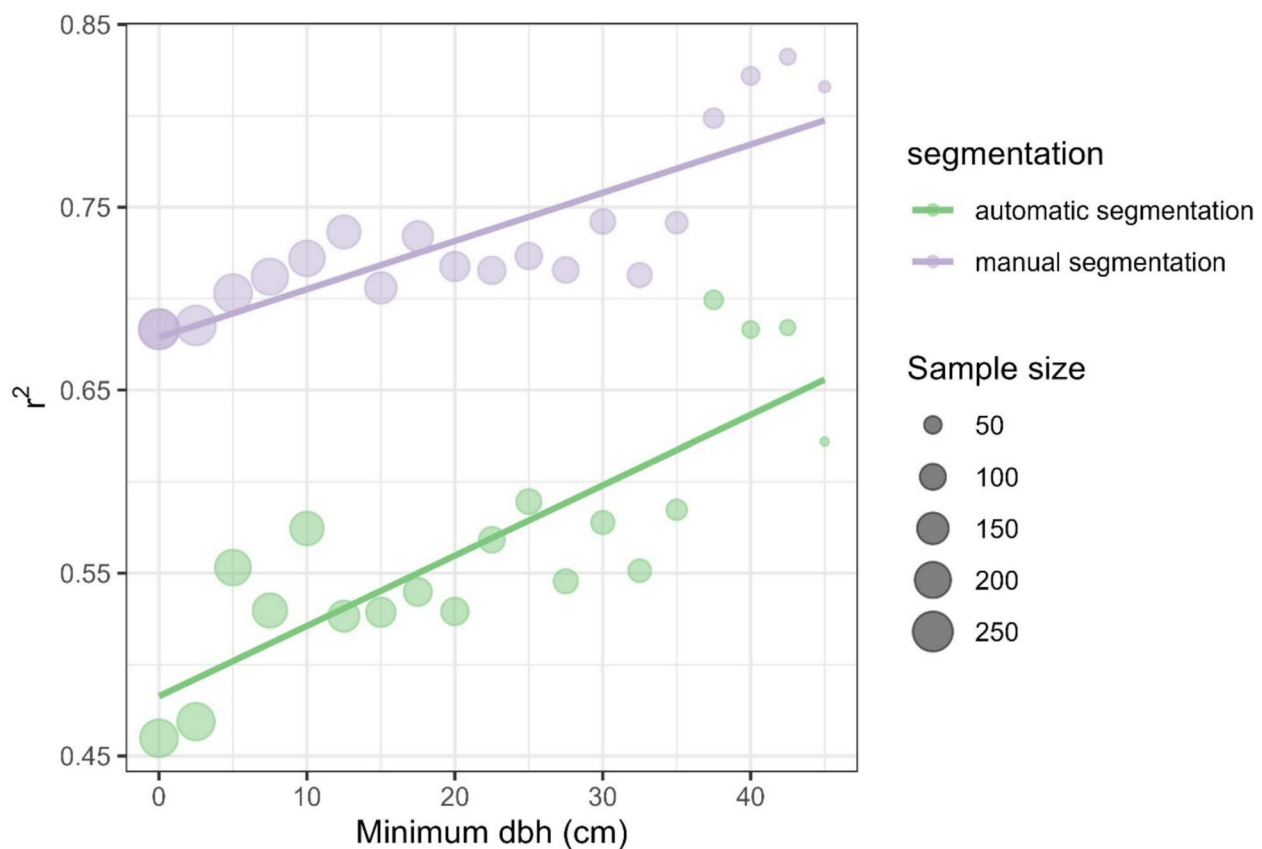


Fig. 8 Relationship between model accuracy (R^2) and the minimum tree size included in the training dataset for random forests model prediction using raw intensity values, comparing results from manually and automatically segmented trees. Accuracy generally improves as smaller trees are excluded from training data

training dataset. In automatically segmented datasets, point clouds of small trees sometimes incorporated non-target vegetation that altered the intensity histogram (Fig. S3), introducing additional noise and decreasing model performance. Therefore, applications of this method should carefully consider the minimum tree size of interest and the choice of segmentation approach, as these factors determine tradeoffs between model accuracy and processing efficiency.

Efficiency relative to ocular measurements

Estimating canopy scorch with terrestrial lidar is more objective relative to ocular measurements and improves efficiency. In our broader study of crown scorch ocular measurements, we completed canopy scorch observations in 70 person-hours and included 1380 trees for a production time of 20 trees h^{-1} . The typical TLS data acquisition and processing rate in our study system is 0.6 ha h^{-1} following the methods herein. The 15-ha study site takes approximately 15 active person-hours for lidar scanning. Following this

step, an additional 10 h of passive time is needed for scan registration; another 15 passive hours are required for pre-processing, tree segmentation, and scorch prediction using a standard desktop computer employing multi-core processing (Intel Core i7-10700 @ 2.90 GHz and 32 GB RAM). Using these steps, we produced scorch estimates for over 6000 pine trees >1 cm. Thus, we collected canopy scorch measurements at a rate of 400 trees h^{-1} including only active time or 150 trees h^{-1} including passive computation time, representing a 20 $\times$ increase compared to ocular measurements, or a 7 $\times$ increase if passive computer processing time is included. Our study area is maintained in an open canopy condition from frequent prescribed burning, and more scanning time may be needed in denser forest types. Wall-to-wall evaluation of canopy scorch over large areas allows for evaluating fire effects at larger scales (e.g., fire effects and their spatial autocorrelation) (Fig. 11B, C). Mobile lidar systems or TLS mounted onto all-terrain vehicles (ATVs) can further increase the rate and scale of fire effects mensuration.

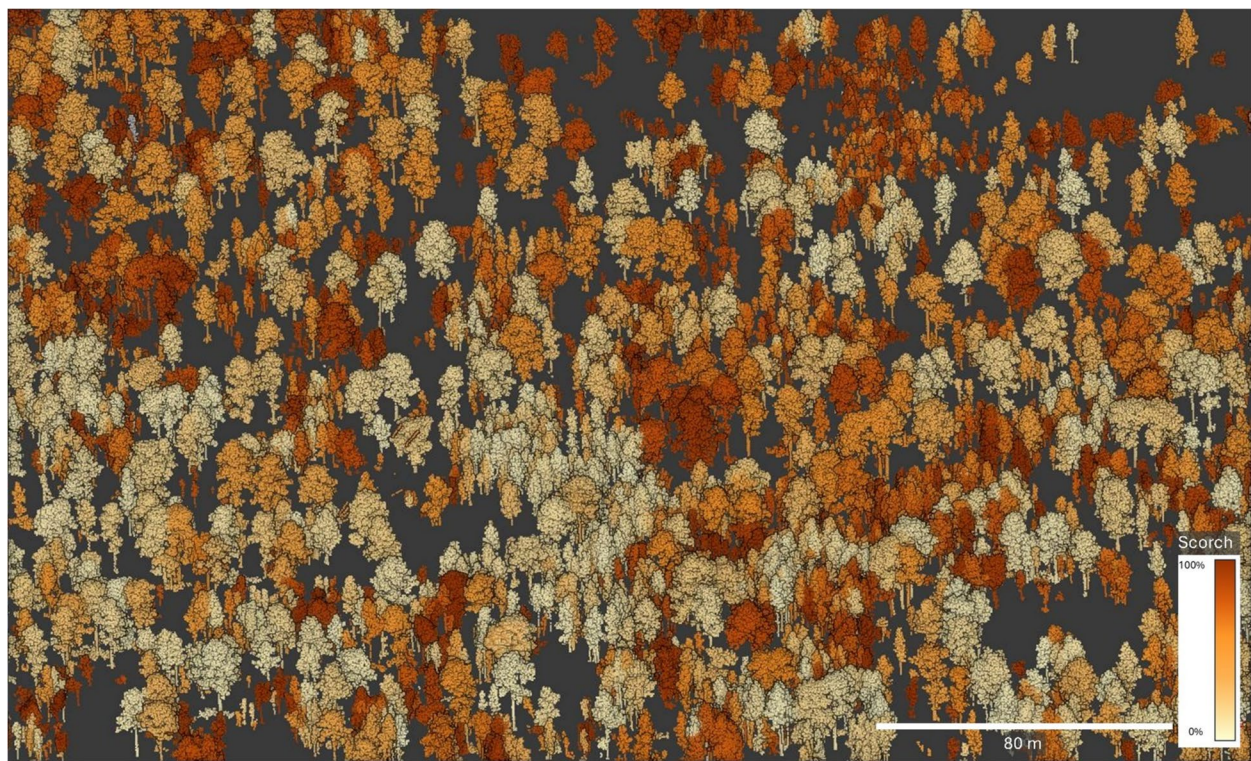


Fig. 9 Three-dimensional rendering of 15-ha study area with automatically segmented *P. palustris* individuals, colored by predicted canopy scorch. Scorch predictions are based on random forests model using return intensity metrics from tree crowns. Predictions were applied to the full point clouds for visualization

Limitations

Terrestrial lidar scanning offers promise for large-scale prediction of crown scorch, but further advances are needed to ensure its practicality, generality, and widespread adoption. First, although lidar systems have long been in use for ecological applications (e.g., Weishampel et al. 1996; Lefsky 1997), techniques for segmenting, extracting, and analyzing individual trees are still nascent and in development (Åkerblom et al. 2016; Cannon 2023; Donager et al. 2021; Sánchez Meador and Donager 2021). Many currently available algorithms for automatic tree segmentation are imperfect and perform best with clear separation between adjacent boles and tree crowns (Sánchez Meador and Donager 2021; Tao et al. 2015; Wielgosz et al. 2024). Thus, we expect near-real-time assessment of post-fire injury to be best suited to forests with widely spaced trees with clear exposed boles. In shrublands dominated by multi-stemmed trunks and interlocking crowns, assigning crown scorch to individual plants using lidar may prove impractical or impossible. In these cases, developing and testing algorithms that predict scorch at the scale of individual lidar returns or voxel-based approaches may be preferred to assigning scorch to individual trees

(e.g., proof-of-concept in Fig. 12). Second, our method produced accurate predictions of crown scorch using a machine learning approach trained by field-based ocular measurements. Field-based ocular measurements used as training data are themselves subjective and variable (Peterson 1985). There may be an upper limit in the accuracy of lidar-derived measurements using this method of model training. Future studies that train models using crown scorch measurements from destructive harvesting and classification of leaves (scorched vs. intact) may improve on the promising results found here. Last, our study focused on a longleaf pine–oak woodland, but mechanisms of crown scorch may vary among individual species anatomy, physiology, and fire behavior (Varner et al. 2021). Thus, we recommend replicate studies in diverse ecosystems to expand the generality of our approach to other ecosystems. Near-real-time assessment of crown scorch following prescribed fires can help fire managers rapidly and accurately assess post-fire injury (and thus mortality), providing valuable feedback on how to adjust ignition techniques to minimize or mitigate injuries to better meet management objectives.



Fig. 10 Demonstration of lidar-derived crown scorch prediction for individual trees automatically segmented within (A) a 30 m by 60 m transect and (B) a 1.44-ha portion of the 15-ha study area

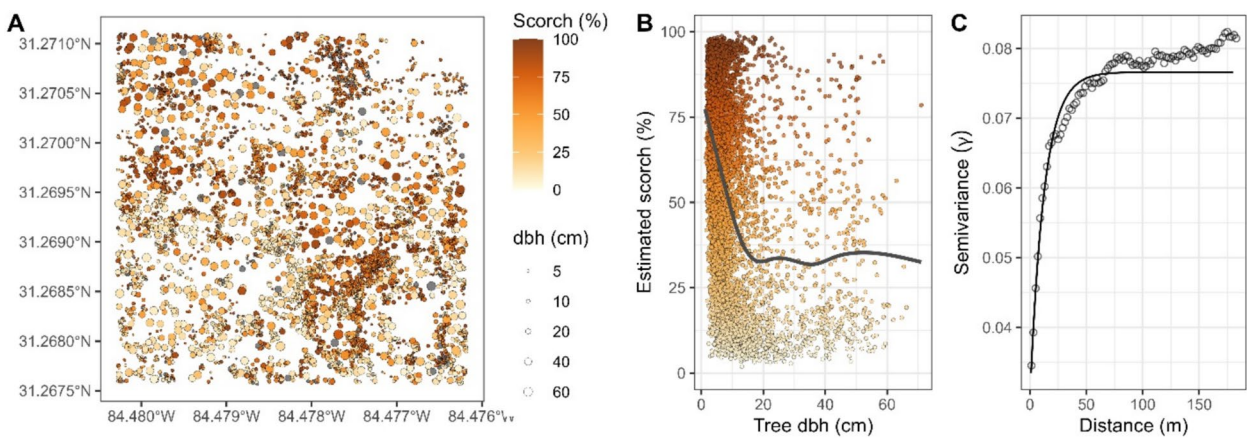


Fig. 11 A Depiction of lidar-derived crown scorch prediction for 15-ha study area showing only *P. palustris* individuals (other species were leaf-off during scanning). Trees shown in gray were not successfully segmented. B Relationship between estimated scorch and tree size for individuals shown in A. Black line represents model fit using loess regression. C Semivariogram of estimates of crown scorch. Black line represents fitted variogram using an exponential model (Pebesma 2004)

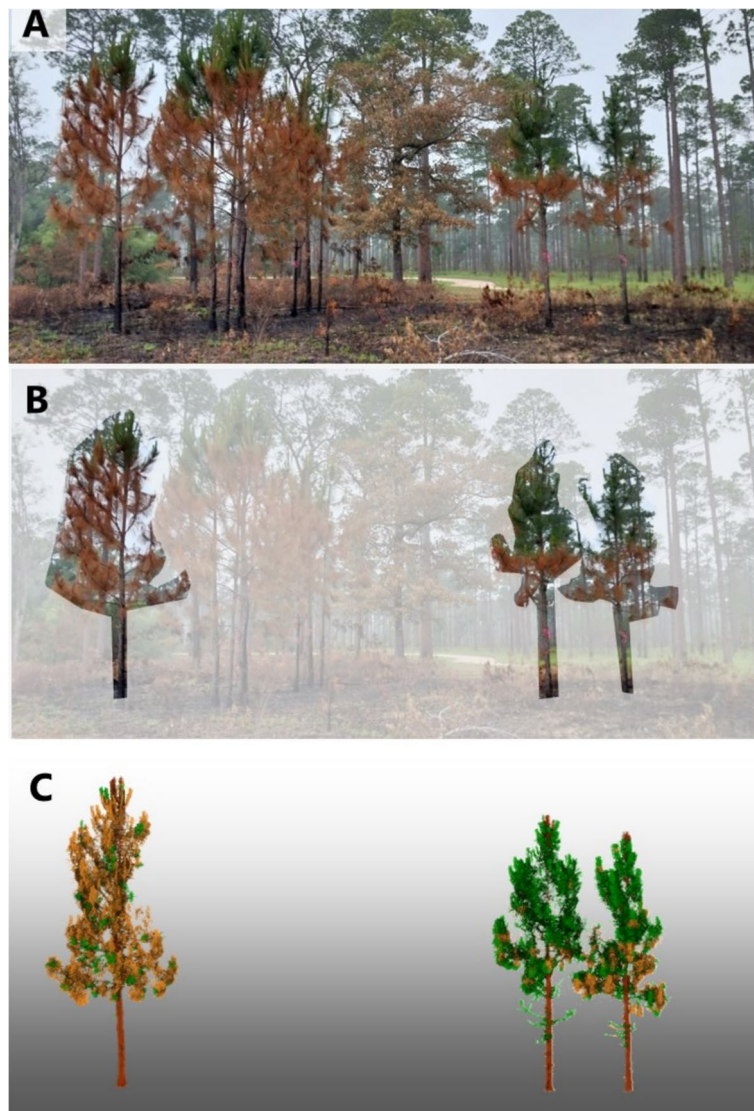


Fig. 12 Proof-of concept demonstrating output of lidar-derived measurement with crown scorch predicted within 15 cm neighborhood to estimate within-tree scrown scorch. **A** Patch of juvenile longleaf pine 2 weeks after prescribed burn, **B** three isolated trees shown with translucent mask over non-focal trees, and **C** rendering of lidar-derived data with points classified as stem (brown), scorched needles (orange), and unscorched needles (green)

Measuring fire effects and tree injury using TLS

Our work focused on detecting the changes in lidar return intensity patterns before and after fire but did not investigate the mechanisms that drive these patterns. We predicted tree-level canopy scorch using TLS, but leaf- or tissue-scale estimates of injury appear feasible using TLS (Fig. 12). Prior studies have used TLS proxies to detect physiological changes in leaves such as water content (Junttila et al. 2018) and chlorophyll content (Saha and Zude-Sasse 2022), and daily to diurnal changes in leaf physiology can be detected (Junttila et al. 2021). The rapid acquisition of TLS and its non-destructive nature

are useful to provide information on leaf physiology at the high spatial and temporal scales necessary to investigate the mechanisms of tree injury. Determining the proximate mechanisms of scorch may enable us to devise methods that isolate these changes, thereby improving our ability to detect tree injury as well as understand the subsequent physiological decline that those specific injuries portend (Hood et al. 2018; O'Brien et al. 2018; Varner et al. 2021).

Rapid and objective measures of fire effects are increasingly needed to aid in decision-making after wildfires, where post-fire salvage decisions are critical. Accurate

measurements of fire effects also guide management with prescribed fires, where evaluation of pre-burn objectives and post-burn planning are necessary elements of adaptive management (Hood et al. 2018). Because crown scorch originates from convective and radiative heating from below the canopy, terrestrial lidar scans are less obscured than ALS techniques from above, whether UAS or another aerial platform. Beyond lidar, other remote sensing tools offer promise to characterize not only brown/green foliage (scorched and unscorched) but greater detail that may better assess more subtle injury to foliage, buds, and underlying branch cambium. Ongoing work with NDVI used alone or combined with lidar could vastly improve estimates of crown injury from individual trees to post-fire stands and landscapes. Multispectral terrestrial lidar units, while currently in development (Takhtkeshha et al. 2024), lag behind single-band terrestrial lidar systems and are not yet widely available. Multispectral terrestrial lidar measurements have been used to estimate changes in leaf moisture content and chlorophyll content (Hakala et al. 2015; Saha and Zude-Sasse 2022; Thomas et al. 2008) and may be useful for unraveling mechanisms of crown scorch through analysis of changes in spectral indexes in tissue-specific regions.

Conclusion

Our results show that return intensity from terrestrial lidar provides a fast, objective, and reliable method for estimating crown scorch. Predictions based on post-burn intensity histograms were produced 20 times faster than field-based ocular assessments. Histograms of return intensity from scorched trees consistently reflected changes in the range of -7.5 and -5.5 dB. Machine learning models using post-burn scans alone accurately predicted scorch. Including pre-burn scans did not improve performance, highlighting the method's practicality for both prescribed fires and wildfires.

To support broader use by the fire science community, we developed the CrownScorchTLS package R package (<https://github.com/jbcannon/CrownScorchTLS>), which generates intensity histograms and applies the best-performing model from this study to predict scorch. Users can also train custom models to accommodate different lidar collection protocols and test the transferability of results to other species or instruments. Post-fire mortality research has made large strides over the past two decades, but the methods to assess injury have lagged. Improved crown scorch measurements, combined with traditional physiological measures of stem water use and leaf physiology (e.g., Ryan and Frandsen 1991; Dickinson and Johnson 2001; O'Brien et al. 2010), should enable

researchers to estimate cumulative stress caused by fires and mark steps in the direction of vastly improving decision-making for land managers using prescribed fire or responding to wildfires. The development and accessibility of the methods herein can help advance this important work.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s42408-025-00420-0>.

Supplementary Material 1

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Authors' contributions

JBC: Methodology, Software, Formal Analysis, Investigation, Resources, Data Curation, Writing—Original Draft, Writing—Review and Editing; NEZ: Methodology, Validation, Investigation, Data Curation, Writing—Review and Editing; AWW: Methodology, Software, Writing—Review and Editing; TMS: Conceptualization, Methodology, Writing—Review and Editing; AJSM: Methodology, Writing—Review and Editing; JMV: Conceptualization, Methodology, Resources, Writing—Original Draft, Writing—Review and Editing.

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Data availability

The authors have created an R package for estimating crown scorch from terrestrial lidar scans that can be found at the following repository: (<https://doi.org/10.5281/zenodo.17380262>) [<https://github.com/jbcannon/CrownScorchTLS>]. Data and code are available for review in the following repository [<https://github.com/jbcannon/CrownScorchTLS-data>] (https://drive.google.com/drive/folders/115unnf2bjl7O-WCxbD6arg_6kynxX0_).

Declarations

Ethics approval and consent to participate

No permissions or licenses were required for this study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Åkerblom, M., P. Raunonen, R. Mäkipää, and M. Kaasalainen. 2016. Automatic tree species recognition with quantitative structure models. *Remote Sensing of Environment* 191:1–12. <https://doi.org/10.1016/j.rse.2016.12.002>.
- Anderson-Teixeira, K. J., S. J. Davies, A. C. Bennett, E. B. Gonzalez-Akre, H. C. Muller-Landau, S. Joseph Wright, K. Abu Salim, A. M. Almeyda Zambrano, A. Alonso, J. L. Baltzer, Y. Bassett, N. A. Bourg, E. N. Broadbent, W. Y. Brockelman, S. Bunyavejchewin, D. F. R. P. Burslem, N. Butt, M. Cao, D. Cardenas, G. B. Chuyong, K. Clay, S. Cordell, H. S. Dattaraja, X. Deng, M. Detto, X. Du, A. Duque, D. L. Erikson, C. E. N. Ewango, G. A. Fischer, C. Fletcher, R. B. Foster, C. P. Giardina, G. S. Gilbert, N. Gunatilleke, S. Gunatilleke, Z. Hao, W. W. Hargrove, T. B. Hart, B. C. H. Hau, F. He, F. M. Hoffman, R. W. Howe, S. P. Hubbell, F. M. Inman-Narahari, P. A. Jansen, M. Jiang, D. J. Johnson, M. Kanzaki, A. R. Kassim, D. Kenfack, S. Kibet, M. F. Kinnaird, L. Korte, K. Kral, J. Kumar, A. J. Larson, Y. Li, X. Li, S. Liu, S. K. Y. Lum, J. A. Lutz, K. Ma, D. M. Maddalena, J. R. Makana, Y. Malhi, T. Marthews, R. Mat Serudin, S. M. McMahon, W. J. McShea, H. R. Memiaghe, X. Mi, T. Mizuno, M. Morecroft, J. A. Myers, V. Novotny, A. A. de Oliveira, P. S. Ong, D. A. Orwig, R. Ostertag, J. den Ouden, G. G. Parker, R. P. Phillips, L. Sack, M. N. Sainge, W. Sang, K. Sri-ngernyuan, R. Sukumar, I. F. Sun, W. Sungpalee, H. S. Suresh, S. Tan, S. C. Thomas, D. W. Thomas, J. Thompson, B. L. Turner, M. Uriarte, R. Valencia, M. I. Vallejo, A. Vicentini, T. Vrška, Xihua Wang, Xugao Wang, G. Weiblen, A. Wolf, H. Xu, S. Yap, and J. Zimmerman. 2015. CTFS-ForestGEO: a worldwide network monitoring forests in an era of global change. *Global Change Biology* 21:528–549. <https://doi.org/10.1111/gcb.12712>.
- Archibald, S., C. E. R. Lehmann, J. L. Gómez-Dans, and R. A. Bradstock. 2013. Defining pyromes and global syndromes of fire regimes. *Proceedings of the National Academy of Sciences of the United States of America* 110:6445–6447. <https://doi.org/10.1073/pnas.1211466110>.
- Arkin, J., N. C. Coops, L. D. Daniels, and A. Plowright. 2023. A novel post-fire method to estimate individual tree crown scorch height and volume using simple RPAS-derived data. *Fire Ecology* 19:17. <https://doi.org/10.1186/s42408-023-00174-7>.
- Arnett, J. T. T. R., N. C. Coops, L. D. Daniels, and R. W. Falls. 2015. Detecting forest damage after a low-severity fire using remote sensing at multiple scales. *International Journal of Applied Earth Observation and Geoinformation* 35:239–246. <https://doi.org/10.1016/j.jag.2014.09.013>.
- Béland, M., D. D. Baldocchi, J.-L. Widowski, R. A. Fournier, and M. M. Verstraete. 2014. On seeing the wood from the leaves and the role of voxel size in determining leaf area distribution of forests with terrestrial LiDAR. *Agricultural and Forest Meteorology* 184:82–97. <https://doi.org/10.1016/j.agrformet.2013.09.005>.
- Breiman, L. 2001. Random forests. *Machine Learning* 45:5–32. <https://doi.org/10.1023/A:1010933404324>.
- Cannon, J. B., 2023. *landecoutils: utilities for the landscape ecology lab for processing terrestrial lidar scans*. <https://github.com/jbcannon/landecoutils>.
- Cansler, C. A., S. M. Hood, P. J. van Mantgem, and J. M. Varner. 2020. A large database supports the use of simple models of post-fire tree mortality for thick-barked conifers, with less support for other species. *Fire Ecology* 16: 25. <https://doi.org/10.1186/s42408-020-00082-0>.
- de Conto, T., 2020. *TreeLS: terrestrial point cloud processing of forest data*. R package version 2.0.4. <https://github.com/tiagodc/TreeLS>.
- De Luca, G., G. Modica, J. M. N. Silva, S. Praticò, and J. M. C. Pereira. 2023. Assessing tree crown fire damage integrating linear spectral mixture analysis and supervised machine learning on Sentinel-2 imagery. *International Journal of Digital Earth* 16:3162–3198. <https://doi.org/10.1080/17538947.2023.2243900>.
- Dickinson, M. B., Johnson, E. A., 2001. Chapter 14 - fire effects on trees, in: Johnson, Edward A., Miyaniishi, K. (Eds.), *Forest fires*. Academic Press, San Diego, pp. 477–525. <https://doi.org/10.1016/B978-012386660-8/50016-7>.
- Donager, J. J., A. J. Sánchez Meador, and R. C. Blackburn. 2021. Adjudicating perspectives on forest structure: how do airborne, terrestrial, and mobile lidar-derived estimates compare? *Remote Sensing* 13:2297. <https://doi.org/10.3390/rs13122297>.
- Engber, E. A., and J. M. Varner. 2012. Predicting Douglas-fir sapling mortality following prescribed fire in an encroached grassland. *Restoration Ecology* 20:665–668. <https://doi.org/10.1111/j.1526-100X.2012.00900.x>.
- Gibson, R., T. Danaher, W. Hehir, and L. Collins. 2020. A remote sensing approach to mapping fire severity in south-eastern Australia using Sentinel 2 and random forest. *Remote Sensing of Environment* 240:111702. <https://doi.org/10.1016/j.rse.2020.111702>.
- Guyette, R., M. Stambaugh, D. Dey, and R.-M. Muzika. 2012. Predicting fire frequency with chemistry and climate. *Ecosystems* 15. <https://doi.org/10.1007/s10021-011-9512-0>.
- Hakala, T., O. Nevalainen, S. Kaasalainen, and R. Mäkipää. 2015. Technical note: multispectral lidar time series of pine canopy chlorophyll content. *Biogeosciences* 12:1629–1634. <https://doi.org/10.5194/bg-12-1629-2015>.
- Harrington, M. G. 1993. Predicting Pinus Ponderosa Mortality from Dormant Season and Growing-Season Fire Injury. *International Journal of Wildland Fire* 3:2:65–72. <https://doi.org/10.1071/wf9930065>.
- Hatton, T. J., N. E. West, and P. S. Johnson. 1986. Relationships of the error associated with ocular estimation and actual total cover. *Journal of Range Management* 39:91. <https://doi.org/10.2307/3899697>.
- Holland, A. M., B. T. Rutledge, S. B. Jack, and J. M. Stober. 2019. The longleaf pine forest: long-term monitoring and restoration of a management dependent ecosystem. *Journal for Nature Conservation* 47:38–50. <https://doi.org/10.1016/j.jnc.2018.11.006>.
- Hood, S. M., J. M. Varner, P. van Mantgem, and C. A. Cansler. 2018. Fire and tree death: understanding and improving modeling of fire-induced tree mortality. *Environmental Research Letters* 13:113004. <https://doi.org/10.1088/1748-9326/a9e934>.
- Jolly, W. M., M. A. Cochrane, P. H. Freeborn, Z. A. Holden, T. J. Brown, G. J. Williamson, and D. M. J. S. Bowman. 2015. Climate-induced variations in global wildfire danger from 1979 to 2013. *Nature Communications* 6:7537. <https://doi.org/10.1038/ncomms8537>.
- Junttila, S., J. Sugano, M. Vastaranta, R. Linnakoski, H. Kaartinen, A. Kukko, M. Holopainen, H. Hyyppä, and J. Hyyppä. 2018. Can leaf water content be estimated using multispectral terrestrial laser scanning? A case study with Norway spruce seedlings. *Frontiers in Plant Science*. 9. <https://doi.org/10.3389/fpls.2018.00299>.
- Junttila, S., T. Hölttä, E. Puttonen, M. Katoh, M. Vastaranta, H. Kaartinen, M. Holopainen, and H. Hyyppä. 2021. Terrestrial laser scanning intensity captures diurnal variation in leaf water potential. *Remote Sensing of Environment* 255:112274. <https://doi.org/10.1016/j.rse.2020.112274>.
- Kursa, M. B., and W. R. Rudnicki. 2010. Feature selection with the boruta package. *Journal of Statistical Software* 36:1–13. <https://doi.org/10.18637/jss.v036.i11>.
- Riegl Laser Measurement Systems GmbH. 2017. *LAS extrabytes implementation in RIEGL software*. <https://data.4tu.nl/file/1aac46fb-7900-4d4c-a099-d2ce354811d2/7ade80c4-aa45-4e87-b887-ee8478c96181>.
- Lefsky, M. A. 1997. *Application of lidar remote sensing to the estimation of forest canopy and stand structure (Ph.D.)*. University of Virginia.
- Liaw, A., and M. Wiener. 2002. Classification and regression by randomForest. *R News* 2:18–22.
- McLauchlan, K. K., P. E. Higuera, J. Miesler, B. M. Rogers, J. Schweitzer, J. K. Shuman, A. J. Tepley, J. M. Varner, T. T. Veblen, S. A. Adalsteinsson, J. K. Balch, P. Baker, E. Battlori, E. Bigio, P. Brando, M. Cattau, M. L. Chipman, J. Coen, R. Crandall, L. Daniels, N. Enright, W. S. Gross, B. J. Harvey, J. A. Hatten, S. Y. Hermann, R. E. Hewitt, L. N. Kobziar, J. B. Landesmann, M. M. Loranty, S. Y. Maezum, L. Mearns, M. Moritz, J. A. Myers, J. G. Pausas, A. F. A. Pellegrini, W. J. Platt, J. Roozeboom, H. Safford, F. Santos, R. M. Scheller, R. L. Sherriff, K. G. Smith, M. D. Smith, and A. C. Watts. 2020. Fire as a fundamental ecological process: research advances and frontiers. *Journal of Ecology* 108:2047–2069. <https://doi.org/10.1111/1365-2745.13403>.
- Moran, C. J., V. Hoff, R. A. Parsons, L. P. Queen, and C. A. Seielstad. 2022. Mapping fine-scale crown scorch in 3D with remotely piloted aircraft systems. *Fire* 5:59. <https://doi.org/10.3390/fire5030059>.
- O'Brien, J. J., Kevin Hiers, J. Mitchell, R. J. Varner, J. M., Mordecai, K. 2010. Acute physiological stress and mortality following fire in a long-unburned longleaf pine ecosystem. *fire ecol* 6:1–12. <https://doi.org/10.4996/fireecology.0602001>.
- O'Brien, J. J., Hiers, J. Varner, C. Hoffman, M. Dickinson, S. Michaletz, L. Loudermilk, and B. Butler. 2018. Advances in mechanistic approaches to quantifying biophysical fire effects. *Current Forestry Reports*. 4. <https://doi.org/10.1007/s40725-018-0082-7>.
- Pebesma, E. J. 2004. Multivariable geostatistics in S: the gstat package. *Computers & Geosciences* 30:683–691. <https://doi.org/10.1016/j.cageo.2004.03.012>.

- Peterson, D. L. 1985. Crown scorch volume and scorch height: estimates of postfire tree condition. *Canadian Journal of Forest Research* 15:596–598. <https://doi.org/10.1139/x85-097>.
- R Core Team. 2023. *R: a language and environment for statistical computing*. <https://www.R-project.org/>.
- Roussel, J. R., D. Auty, N. C. Coops, P. Tompalski, T. R. H. Goodbody, A. Sánchez Meador, J. F. Bourdon, F. De Boissieu, and A. Achim. 2020. LidR: an R package for analysis of airborne laser scanning (ALS) data. *Remote Sensing of Environment* 251:112061.
- Ryan, K., and W. Frandsen. 1991. Basal injury from smoldering fires in mature *Pinus ponderosa* Laws. *International Journal of Wildland Fire* 1:107. <https://doi.org/10.1071/WF9910107>.
- Ryan, K. C., Knapp, E. E., Varner, J. M. 2013. Prescribed fire in North American forests and woodlands: history, current practice, and challenges. *Frontiers in Ecology and the Environment* 11. <https://doi.org/10.1890/120329>.
- Saha, K. K., and M. Zude-Sasse. 2022. Estimation of chlorophyll content in banana during shelf life using LiDAR laser scanner. *Postharvest Biology and Technology* 192:112011. <https://doi.org/10.1016/j.postharvbio.2022.112011>.
- Sánchez Meador, A., Donager, J. 2021. *spanner: utilities to support landscape-, forest-, and tree-related data analysis*. <https://github.com/biom3trics/spanner>.
- NRCS Soil Survey Staff. 2023. *Soil Survey Geographic (SSURGO) Database*. Natural Resources Conservation Service, United States Department of Agriculture. Available online at <https://sdmdataaccess.sc.egov.usda.gov>.
- Takhtkeshha, N., G. Mandlbürger, F. Remondino, and J. Hyypä. 2024. Multispectral light detection and ranging technology and applications: a review. *Sensors* 24:1669. <https://doi.org/10.3390/s24051669>.
- Tao, S., F. Wu, Q. Guo, Y. Wang, W. Li, B. Xue, X. Hu, P. Li, D. Tian, C. Li, H. Yao, Y. Li, G. Xu, and J. Fang. 2015. Segmenting tree crowns from terrestrial and mobile LiDAR data by exploring ecological theories. *ISPRS Journal of Photogrammetry and Remote Sensing* 110:66–76. <https://doi.org/10.1016/j.isprsjprs.2015.10.007>.
- Thomas, V., P. Treitz, J. H. McCaughey, T. Noland, and L. Rich. 2008. Canopy chlorophyll concentration estimation using hyperspectral and lidar data for a boreal mixedwood forest in northern Ontario, Canada. *International Journal of Remote Sensing* 29:1029–1052. <https://doi.org/10.1080/0143160701281023>.
- Varner, J. M., S. M. Hood, Doug.P. Aubrey, K. Yedinak, J. K. Hiers, W. M. Jolly, T. M. Shearman, J. K. McDaniel, J. J. O'Brien, and E. M. Rowell. 2021. Tree crown injury from wildland fires: causes, measurement and ecological and physiological consequences. *New Phytologist* 231:1676–1685. <https://doi.org/10.1111/nph.17539>.
- Weishampel, J. F., K. J. Ranson, and D. J. Harding. 1996. Remote sensing of forest canopies. *Selbyana* 17:6–14.
- Wielgosz, M., S. Puliti, B. Xiang, K. Schindler, and R. Astrup. 2024. Segmen-tAnyTree: a sensor and platform agnostic deep learning model for tree segmentation using laser scanning data. *Remote Sensing of Environment* 313:114367. <https://doi.org/10.1016/j.rse.2024.114367>.
- Wing, B. M., M. W. Ritchie, K. Boston, W. B. Cohen, A. Gitelman, and M. J. Olsen. 2012. Prediction of understory vegetation cover with airborne lidar in an interior ponderosa pine forest. *Remote Sensing of Environment* 124:730–741. <https://doi.org/10.1016/j.rse.2012.06.024>.
- Xu, T., L. Xu, B. Yang, X. Li, and J. Yao. 2017. Terrestrial laser scanning intensity correction by piecewise fitting and overlap-driven adjustment. *Remote Sensing* 9:1090. <https://doi.org/10.3390/rs9111090>.
- Zhang, G., and Y. Lu. 2012. Bias-corrected random forests in regression. *Journal of Applied Statistics* 39:151–160. <https://doi.org/10.1080/02664763.2011.578621>.
- Zhu, X., T. Wang, A. K. Skidmore, R. Darvishzadeh, K. O. Niemann, and J. Liu. 2017. Canopy leaf water content estimated using terrestrial LiDAR. *Agricultural and Forest Meteorology* 232:152–162. <https://doi.org/10.1016/j.agrformet.2016.08.016>.

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